CLASS 9 · PART TEST-1 · STUDENT PERFORMANCE REPORT

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Campus HYD/Nallakunta DD Colony
JEE Mains 08 Oct 2026
JEE Advanced 09 Oct 2026

JEE Mains

08 Oct 2026 · 74 questions

146 / 296 — **49.3%**

202	137	14/79	33
ALL-INDIA	STATE	CAMPUS	PROGRAM

Weekly-test average **153.5 / 300** (51.2%) → this test **49.3% ▼ 1.9**

150 marks lost: 70 on 14 wrong answers (incl. 14 negative) · 80 on 20 skipped

JEE Advanced

09 Oct 2026 · 50 questions

87 / 176 — **49.4%**

87	63	8/78	11†
ALL-INDIA	STATE	CAMPUS	PROGRAM

Weekly-test average **72.2 / 180** (40.1%), 5 of 6 tests → this test **49.4% ▲ 9.3**

89 marks lost: 57 on 15 wrong answers (incl. 4 negative) · 26 on 8 skipped · 6 on partly-correct answers

1 Subject scorecard

Your marks in each subject, against the 79 students of this program on campus and against your own weekly-test average.

SUBJECT	MARKS	SCORE VS CAMPUS	SUBJECT AIR	CAMPUS AVG / BEST	WEEKLY AVG	CHANGE	RIGHT / WRONG / SKIP
JEE Mains 146 / 296 · 49.3%							
Maths	**60**/100	60%	**209**	43.1 / 95	**68.2**/100 68.2%	▼ **8.2**	17 / 8 / 0
Physics	**14**/100	14%	**837**	21.9 / 90	**47.8**/100 47.8%	▼ **33.8**	4 / 2 / 19
Chemistry	**72**/96	75%	**23**	41.4 / 81	**37.1**/100 37.1%	▲ **37.9**	19 / 4 / 1
JEE Advanced 87 / 176 · 49.4%							
Maths	**39**/60	65%	**130**	24.5 / 60	**27.4**/60 45.6%	▲ **19.4**	12 / 4 / 1
Physics	**13**/60	22%	**498**	12.3 / 56	**22.5**/60 37.6%	▼ **15.9**	4 / 6 / 7
Chemistry	**35**/56	62%	**23**	15.8 / 37	**22.3**/60 37.1%	▲ **25.4**	11 / 5 / 0

Bar = your score % | campus average | best on campus Change = this test % − weekly-test average %

What this report says

1. **Chemistry** is your strongest JEE Mains subject: 72/96 (75%), All-India subject rank 23.
2. **Physics** needs the most work in JEE Mains: 14/100 (14%), subject rank 837; you lost 21 of its 25 questions.
3. **Maths** is your strongest JEE Advanced subject: 39/60 (65%), All-India subject rank 130.
4. **Physics** needs the most work in JEE Advanced: 13/60 (22%), subject rank 498; you lost 13 of its 17 questions.
5. Biggest fall against your weekly-test average: **JEE Mains Physics** – 14% here against 48% on average.
6. Biggest improvement: **JEE Mains Chemistry** – 75% here against 37% on average.
7. **2 questions** you lost were answered correctly by 70%+ of the batch's top 100 – about **10 marks** within easy reach (Mains Q23, Q52).
8. Negative marking cost you **14 in JEE Mains and 4 in JEE Advanced**.
9. You left **20 in JEE Mains and 8 in JEE Advanced** questions blank, and answered 14 in JEE Mains and 15 in JEE Advanced wrong – both are marks to recover.

JEE Mains – question by question

146 / 296 · 49.3%
08 Oct 2026

2 Every question at a glance

Right · Wrong · Skipped · Deleted (no marks) · Easy miss · Within reach · Tough one

Maths	1	2	3	4	5	**6**	7	**8**	9	10	11	12	**13**	14	15	16	**17**	**18**	19	**20**	**21**	22	**23**	24	25
Physics	**26**	**27**	**28**	**29**	**30**	**31**	**32**	**33**	**34**	**35**	**36**	**37**	**38**	39	**40**	**41**	**42**	**43**	44	**45**	46	47	**48**	**49**	**50**
Chemistry	51	**52**	53	54	55	56	57	~~58~~	59	60	61	62	63	64	65	66	**67**	68	69	70	71	**72**	73	**74**	**75**

3 Questions you lost – what to fix first

Every question you got wrong or skipped, one table per subject, easiest marks first. **Top 100** = the batch's top 100 across all campuses; **Campus** = all 79 students of this program on campus.

Easy miss 70%+ of the top 100 got it right – easiest marks to win back · **Within reach** 40–69% · **Tough one** below 40%

Maths

8 of 25 questions lost · **40** marks · ● 1 easy miss · ● 4 within reach · ● 3 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q23	Easy miss	**Inverse Trigonometric Functions**	Numerical · +4 / −1	Wrong	5	71%	48%
Q13	Within reach	**Trigonometric Equations** – Solutions of tan x + sec x = 2 cos x in [0, 2π]	Single correct · +4 / −1	Wrong	5	66%	58%
Q6	Within reach	**Inverse Trigonometric Functions**	Single correct · +4 / −1	Wrong	5	58%	35%
Q18	Within reach	**Sequences & Series**	Single correct · +4 / −1	Wrong	5	50%	49%
Q21	Within reach	**Trigonometric Equations**	Numerical · +4 / −1	Wrong	5	50%	33%
Q8	Tough one	**Functions**	Single correct · +4 / −1	Wrong	5	37%	27%
Q17	Tough one	**Inverse Trigonometric Functions**	Single correct · +4 / −1	Wrong	5	34%	35%
Q20	Tough one	**Inverse Trigonometric Functions** – Composite inverse-trig function – counting solutions	Single correct · +4 / −1	Wrong	5	19%	13%

Physics

21 of 25 questions lost · **86** marks · ● 9 within reach · ● 12 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q34	Within reach	**Work, Energy & Power**	Single correct · +4 / −1	Skipped	4	66%	47%
Q40	Within reach	**Laws of Motion**	Single correct · +4 / −1	Skipped	4	63%	46%
Q50	Within reach	**Momentum & Collisions**	Numerical · +4 / −1	Skipped	4	60%	41%
Q29	Within reach	**Friction**	Single correct · +4 / −1	Skipped	4	59%	38%
Q41	Within reach	**Laws of Motion**	Single correct · +4 / −1	Skipped	4	50%	39%
Q37	Within reach	**Laws of Motion** – Tension in a pulley–block system	Single correct · +4 / −1	Skipped	4	49%	43%
Q48	Within reach	**Momentum & Collisions**	Numerical · +4 / −1	Skipped	4	47%	34%
Q30	Within reach	**Circular Motion**	Single correct · +4 / −1	Skipped	4	43%	27%
Q33	Within reach	**Laws of Motion** – Tangential acceleration on a frictionless curved track	Single correct · +4 / −1	Skipped	4	43%	25%
Q36	Tough one	**Laws of Motion**	Single correct · +4 / −1	Skipped	4	39%	24%
Q27	Tough one	**Circular Motion** – Coins on a rotating disc – friction limit at each radius	Single correct · +4 / −1	Skipped	4	37%	23%
Q49	Tough one	**Laws of Motion**	Numerical · +4 / −1	Skipped	4	35%	34%
Q35	Tough one	**Circular Motion** – Power delivered by the centripetal force	Single correct · +4 / −1	Wrong	5	34%	18%
Q32	Tough one	**Circular Motion** – Blocks on a rotating frame – ω when sliding starts	Single correct · +4 / −1	Skipped	4	33%	28%
Q31	Tough one	**Work, Energy & Power** – KE vs height for a stone thrown up from a cliff	Single correct · +4 / −1	Wrong	5	32%	10%
Q42	Tough one	**Work, Energy & Power** – Change in PE of a ball thrown from a tower	Single correct · +4 / −1	Skipped	4	32%	15%
Q26	Tough one	**Momentum & Collisions** – KE loss in head-on collision from relative velocities	Single correct · +4 / −1	Skipped	4	31%	22%
Q38	Tough one	**Work, Energy & Power** – Work along different paths; conservative or not	Single correct · +4 / −1	Skipped	4	25%	13%
Q43	Tough one	**Laws of Motion** – Tension in a spring pulled by 5 N at each end	Single correct · +4 / −1	Skipped	4	24%	10%
Q28	Tough one	**Friction** – Friction vs incline angle graph (angle of repose)	Single correct · +4 / −1	Skipped	4	21%	11%
Q45	Tough one	**Work, Energy & Power** – Speed on hitting the ground – energy conservation	Single correct · +4 / −1	Skipped	4	18%	6%

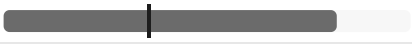
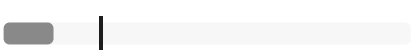
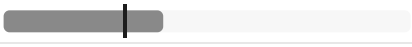
Chemistry

5 of 24 questions lost · **24** marks · ● 1 easy miss · ● 4 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q52	Easy miss	**Isomerism**	Single correct · +4 / −1	Wrong	5	73%	70%
Q72	Tough one	**Hydrocarbons**	Numerical · +4 / −1	Wrong	5	28%	23%
Q67	Tough one	**General Organic Chemistry** – Structure and hybridisation of the methyl carbocation	Single correct · +4 / −1	Wrong	5	26%	28%
Q74	Tough one	**General Organic Chemistry** – Compounds giving a positive bicarbonate test	Numerical · +4 / −1	Skipped	4	20%	11%
Q75	Tough one	**General Organic Chemistry** – Bases stronger than aniline	Numerical · +4 / −1	Wrong	5	11%	9%

4 Chapter scorecard

Your marks on each chapter's questions against the campus average for the same questions. Weakest chapters first within each subject.

CHAPTER	QS	YOUR MARKS	YOU VS CAMPUS AVERAGE	CAMPUS AVG	QUESTIONS YOU LOST
Maths					
Sequences & Series	1	**-1**/4		45%	**Q18**
Inverse Trigonometric Functions	10	**20**/40		48%	**Q6**, **Q17**, **Q20**, **Q23**
Trigonometric Equations	7	**18**/28		44%	**Q13**, **Q21**
Functions	7	**23**/28		35%	**Q8**
Physics					
Work, Energy & Power	5	**-1**/20		9%	**Q31**, **Q34**, **Q38**, **Q42**, **Q45**
Friction	2	**0**/8		17%	**Q28**, **Q29**
Momentum & Collisions	3	**0**/12		25%	**Q26**, **Q48**, **Q50**
Laws of Motion	8	**4**/32		23%	**Q33**, **Q36**, **Q37**, **Q40**, **Q41**, **Q43**, **Q49**
Circular Motion	7	**11**/28		30%	**Q27**, **Q30**, **Q32**, **Q35**
Chemistry					
General Organic Chemistry	10	**26**/40		32%	**Q67**, **Q74**, **Q75**
Hydrocarbons	4	**11**/16		42%	**Q72**
Isomerism	10	**35**/40		55%	**Q52**

Bar = your % of the chapter's marks | campus average

5 JEE Mains · Maths – questions to revisit

8 questions · easy misses first
cropped from the question paper

Q23 Inverse Trigonometric Functions

JEE Mains · Maths · Numerical · marking +4 / −1 · marks lost 5
Top 100: **71%** got it right · Campus: **48%** (38 of 79)

Easy miss
You: Wrong

23. $\cos^2\left(\tan^{-1}\left(\sin\left(\cot^{-1}(3)\right)\right)\right) = ...$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q13 Trigonometric Equations – Solutions of tan x + sec x = 2 cos x in [0, 2π]

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **66%** got it right · Campus: **58%** (46 of 79)

Within reach
You: Wrong

13. Number of solution(s) of $\tan x + \sec x = 2\cos x$ in $[0, 2\pi]$ is

A) 1 B) 2 C) 3 D) 4

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q6 Inverse Trigonometric Functions

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **58%** got it right · Campus: **35%** (28 of 79)

Within reach
You: Wrong

6. $\tan^{-1}(2) + \tan^{-1}(3) = ...$

A) $\left(-\frac{\pi}{4}\right)$ B) $\frac{\pi}{4}$ C) $\frac{3\pi}{4}$ D) $-\frac{3\pi}{4}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q18

Sequences & Series

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **50%** got it right · Campus: **49%** (39 of 79)

Within reach

You: Wrong

18. Let $f(n)=1^3+2^3+3^3+4^3+.....+n^3$ then the sum $S=1^3+3^3+5^3+.....+(2n-1)^3$ can be written as :

A) $f(2n)+8f(n)$

B) $8f(2n)+f(n)$

C) $f(2n)-8f(n)$

D) $8f(n)-f(2n)$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q21

Trigonometric Equations

JEE Mains · Maths · Numerical · marking +4 / −1 · marks lost 5

Top 100: **50%** got it right · Campus: **33%** (26 of 79)

Within reach

You: Wrong

21. Number of solutions of $(\sin x)^{2026}+(\cos x)^{2026}=1$ if $x\in[-10\pi,10\pi]$ is ……..

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q8

Functions

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **37%** got it right · Campus: **27%** (21 of 79)

Tough one

You: Wrong

8. Let $f(x)$ denote the square of sum of digits of $n\in N$. Let $f_2(x)=f(f(x)), f_3(x)=f(f(f(x)))$ and so on. Then, $\dfrac{f_{2026}(2011)-f_{2023}(2011)}{f_{2024}(2011)-f_{2025}(2011)}=$ ……

A) 0

B) -1

C) 1

D) 2

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q17

Inverse Trigonometric Functions
JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **34%** got it right · Campus: **35%** (28 of 79)

Tough one
You: Wrong

17. Let $\left(\sin^{-1}x\right)^2+\left(\cos^{-1}x\right)^2=k\left(where\ \ x\in[-1,1]\right)$ then k cannot be

A) $\frac{\pi^2}{8}$ B) $\frac{\pi^2}{3}$ C) $\frac{\pi^2}{9}$ D) π^2

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q20

Inverse Trigonometric Functions – Composite inverse-trig function – counting solutions
JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **19%** got it right · Campus: **13%** (10 of 79)

Tough one
You: Wrong

20. Consider the functions $rage(x)=\sin^{-1}\sqrt{x}+e^{\left(\tan^{-1}\left(\sqrt{x-1}\right)\right)^2}+\left(\cos^{-1}\left(\sqrt{x}\right)\right)^3$ and $bait(x)=x^2+x+1$ within their respective domains. Then the number of solutions of $rage\left(bait(x)\right)<3$ is/are ……..

A) 0 B)1 C)2 D)3

Why did I lose this question? (write the reason and the correct approach)

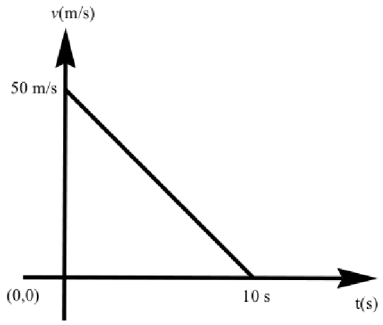
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q34

Work, Energy & Power
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4
Top 100: **66%** got it right · Campus: **47%** (37 of 79)

Within reach
You: Skipped

34. The velocity-time graph of a particle of mass 10kg is shown in fig. the net workdone on the particle in the first two seconds of the motion is



A) 10,000J B) -12,000J C) -4500J D) 6000J

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q40

Laws of Motion
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4
Top 100: **63%** got it right · Campus: **46%** (36 of 79)

Within reach
You: Skipped

40. Statement – I: Force required to move a body uniformly along straight line is can be zero.

Statement – II: The force required to move a body uniformly along the circle is zero.

A) Statement – I is true , Statement – II false

B) Both Statements are true

C) Both Statements are false

D) Statement – I false , Statement – II true

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q50

Momentum & Collisions

JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 4

Top 100: **60%** got it right · Campus: **41%** (32 of 79)

Within reach

You: Skipped

50. When kinetic energy of a body becomes 64 times of its original value, the percentage increase in momentum of body will be ……%

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q29

Friction

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **59%** got it right · Campus: **38%** (30 of 79)

Within reach

You: Skipped

29. An insect is crawling up on a concave surface of a fixed hemispherical bowl of radius R. If the co-efficient of friction is $\frac{1}{3}$ then the height up to which an insect can crawl is nearly $\left[\sqrt{10} \approx 3.16\right]$

A) 5 % of R B) 6 % of R C) 6.5% of R D) 7.5 % of R

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q41

Laws of Motion

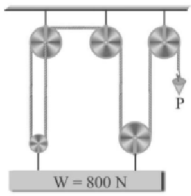
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **50%** got it right · Campus: **39%** (31 of 79)

Within reach

You: Skipped

41. For the arrangement of pulley shown in fig, the effort (P) required to raise the given load (w). assume efficiency of the system as 80%



A) 200N B) 250N C) 500N D) 1000N

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q37 — Laws of Motion – Tension in a pulley-block system

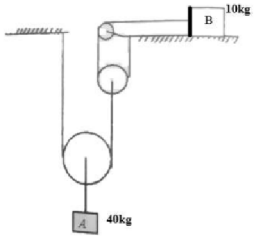
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **49%** got it right · Campus: **43%** (34 of 79)

Within reach

You: Skipped

37. Fig shows two blocks A and B connected to an ideal pulley string system. In this system when bodies are released then $\left(g = 10m/s^2\right)$(Mass of a is 40Kg and B is 10Kg on smooth surface). The tension in the string connecting the block B is



A) $20N$ B) $80N$ C) $60N$ D) $40N$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q48 — Momentum & Collisions

JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 4

Top 100: **47%** got it right · Campus: **34%** (27 of 79)

Within reach

You: Skipped

48. An artillery piece of mass 10kg fires a shell of mass 2kg horizontally. Instantaneously after the firing, the ratio of kinetic energy of shell and that of artillery is …….

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q30

Circular Motion

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **43%** got it right · Campus: **27%** (21 of 79)

Within reach

You: Skipped

30. When a bucket containing water is rotated fast in a vertical circle of radius R, the water in the bucket doesn't spill provided that bucket is

A) whirled around with maximum speed of $\sqrt{2gR}$

B) whirled around with minimum speed of $\sqrt{\frac{gR}{2}}$

C) having a rpm of $n = \sqrt{900g / \pi^2 R}$

D) having a rpm of $n = \sqrt{3600g / \pi^2 R}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q33

Laws of Motion – Tangential acceleration on a frictionless curved track

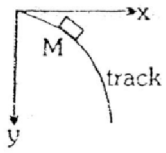
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **43%** got it right · Campus: **25%** (20 of 79)

Within reach

You: Skipped

33. A man M slides down a curved frictionless track, starting from rest. The curve obeys equation $y = \frac{x^2}{2}$. The tangential acceleration of mass is



A) g B) $\frac{gx}{\sqrt{x^2+4}}$ C) $\frac{g}{2}$ D) $\frac{gx}{\sqrt{x^2+1}}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q36 Laws of Motion

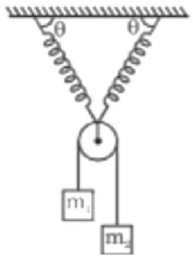
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **39%** got it right · Campus: **24%** (19 of 79)

Tough one

You: Skipped

36. Two identical ideal springs of spring constant (K) 1000N/m as connected by an ideal pulley as shown and system is arranged in vertical plane. Δt equilibrium θ is 60° and masses m_1 and m_2 are 2kg and 3kg respectively. Then elongation in each spring when $\theta = 60^o$ is $\left(g = 10m/s^2\right)$



A) $x = 1.6cm$ B) $x = 4.8cm$ C) $x = 4.8\sqrt{3}$ D) $x = 1.6\sqrt{3}cm$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q27 Circular Motion – Coins on a rotating disc – friction limit at each radius

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **37%** got it right · Campus: **23%** (18 of 79)

Tough one

You: Skipped

27. A disc revolves with a speed of 100/3 rev/min and has radius of 15cm. Two coins of same mass as disc are placed at 4 cm and 14cm away from the centre of disc. If the coefficient of friction between the coins and the disc is 0.15, which of the two coins will revolve with disc?
A) both the coins will revolve
B) No coin will revolve with disc
C) coin at 4cm will revolve with disc
D) coin at 14cm will revolve with disc

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q49 Laws of Motion

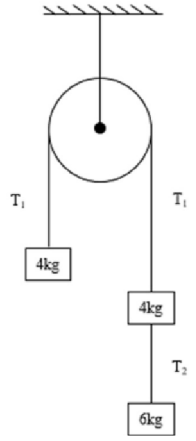
JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 4

Top 100: **35%** got it right · Campus: **34%** (27 of 79)

Tough one

You: Skipped

49. Assume pulley and string are ideal then find $\frac{T_2}{T_1} = \ldots\ldots\ldots$



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q35 Circular Motion – Power delivered by the centripetal force

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **34%** got it right · Campus: **18%** (14 of 79)

Tough one

You: Wrong

35. A particle of mass m is moving in a circular path of constant radius r such that its centripetal acceleration changing with time 't' as $a_c = k^2t^2$, where K is a constant. The power delivered to the particle by centripetal force acting on it is

A) zero B) mk^2rt C) mk^2r^2t D) $mk^2\sqrt{rt}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q32 **Circular Motion – Blocks on a rotating frame – ω when sliding starts**

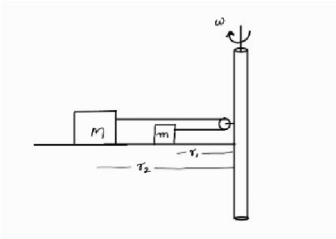
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **33%** got it right · Campus: **28%** (22 of 79)

Tough one

You: Skipped

32. Two blocks of masses m and M are connected by a chord passing around a frictionless pulley which is attached to a rotating frame, which rotates about a vertical axis with angular velocity ω. If co-efficient of friction between two masses and the surface be μ_1 and μ_2 respectively. The value of ω at which block starts sliding radially $(M > m)$



A) $\omega = \left[g \left[\frac{\mu_1 M + \mu_2 m}{m r_1 - M r_1} \right] \right]^{1/2}$

B) $\omega = \left[g \left[\frac{\mu_1 m + \mu_2 M}{M r_2 - m r_1} \right] \right]^{1/2}$

C) $\omega = \left[g \left[\frac{\mu_1 m - \mu_2 M}{M r_2 - m r_1} \right] \right]^{1/2}$

D) $\omega = \left[g \left[\frac{\mu_1 M - \mu_2 m}{M r_2 - m r_1} \right] \right]^{1/2}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q31

Work, Energy & Power – KE vs height for a stone thrown up from a cliff

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

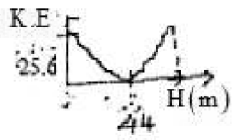
Top 100: **32%** got it right · Campus: **10%** (8 of 79)

Tough one

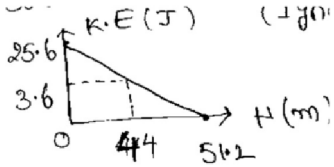
You: Wrong

31. A stone of mass 50g is thrown vertically upwards with a velocity of 12m/s from edge of a cliff 44m high. Identify kinetic energy of stone versus height from bottom of cliff $\left(g = 10m/s^2\right)$ (ignore air resistance)

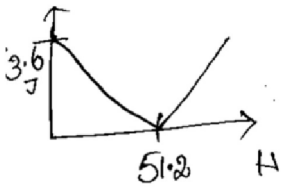
A)



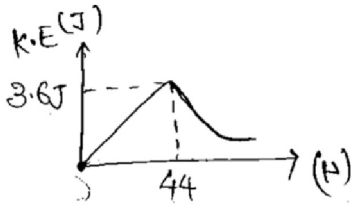
B)



C)



D)



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q42

Work, Energy & Power – Change in PE of a ball thrown from a tower

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **32%** got it right · Campus: **15%** (12 of 79)

Tough one

You: Skipped

42. At t =0 sec a 1kg ball is thrown from top to a tall tower with velocity $\vec{V} = \left(8\hat{i} + 9\hat{j}\right)$ m/s. The change in potential energy of ball – earth system between $t = 0$ *to* $t = 5$sec is

$\left(g = 10m/s^2\right)$

A) 800J B) 1536J C) 2100J D) -800J

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q26

Momentum & Collisions – KE loss in head-on collision from relative velocities

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **31%** got it right · Campus: **22%** (17 of 79)

Tough one

You: Skipped

26. Two bodies of masses 3 kg and 2 kg collide head on. Their relative velocities before and after collision are 15m/s and 5 m/s respectively. The loss of kinetic energy of system is

A) 120J B) 100J C) 80J D) 240J

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q38

Work, Energy & Power – Work along different paths; conservative or not

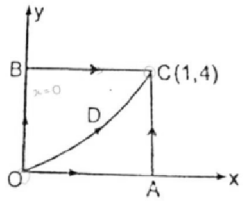
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **25%** got it right · Campus: **13%** (10 of 79)

Tough one

You: Skipped

38. A particle is moved along the different paths OAC, OBC and ODC as shown in the fig. path ODC is parabola, $y = 4x^2$. The work done by a force $\vec{F} = xy\hat{i} + x^2 y\hat{j}$ on the particle along



Column – I	Column – II
a)Path OAC	P) 2J
b) Path OBC	Q) $\frac{19}{3}$J
c) Path ODC	R) Conservative
d) Force is	S) 8J
	T) Non- conservative

A) $a \to Q, b \to S, c \to P, d \to T$ B) $a \to S, b \to P, c \to Q, d \to T$

C) $a \to S, b \to Q, c \to P, d \to T$ D) $a \to S, b \to Q, c \to S, d \to T$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q43

Laws of Motion – Tension in a spring pulled by 5 N at each end

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **24%** got it right · Campus: **10%** (8 of 79)

Tough one

You: Skipped

43. The tension in the massless spring is



A) zero B) 10N C) 2.5N D) 5N

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q28

Friction – Friction vs incline angle graph (angle of repose)

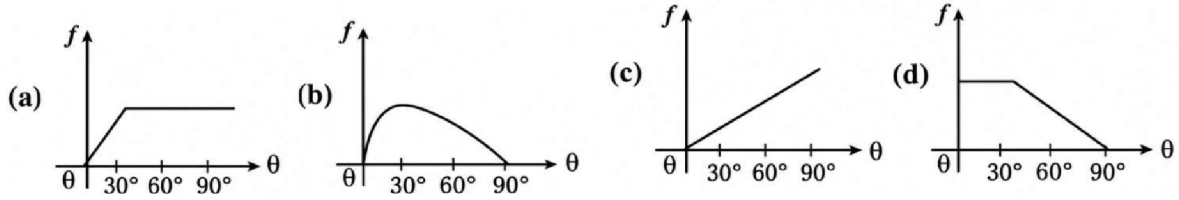
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **21%** got it right · Campus: **11%** (9 of 79)

Tough one

You: Skipped

28. A block is placed on a rough plane whose inclination to the horizontal (θ) can be varied. The angle of repose is 30°. The graph that correctly indicate the variation of frictional force (f) between block and the plane with θ is



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q45 **Work, Energy & Power** – **Speed on hitting the ground – energy conservation**
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4
Top 100: **18%** got it right · Campus: **6%** (5 of 79)

Tough one
You: Skipped

45. A cricket bowler releases the ball in two different ways

i) giving it only horizontal velocity and

ii) giving it horizontal velocity and small downward velocity

The speeds at the time release is same. Both are released at a height H from ground then which one will have greater speed when ball hits the ground?(Neglect air resistance)

A) Ist way B) IInd way C) same in two ways D) None of the above

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Mains · Chemistry – questions to revisit

5 questions · easy misses first
cropped from the question paper

Q52 Isomerism

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **73%** got it right · Campus: **70%** (55 of 79)

Easy miss
You: Wrong

52. The number of structurally isomeric tribromo derivatives possible for benzene are :
A) 2 B) 3 C) 4 D) 5

Why did I lose this question? (write the reason and the correct approach)

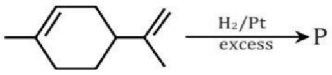
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q72 Hydrocarbons

JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 5
Top 100: **28%** got it right · Campus: **23%** (18 of 79)

Tough one
You: Wrong

72. How many products will be formed for given reaction, including stereo isomers



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q67 General Organic Chemistry – Structure and hybridisation of the methyl carbocation

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **26%** got it right · Campus: **28%** (22 of 79)

Tough one
You: Wrong

67. Wrong statement regarding methyl carbonium ion $\overset{\oplus}{C}H_3$

A) It is sp^2 hybridized

B) Vacant orbital is sp^2 hybridized

C) vacant orbital is perpendicular to molecular planar and is pure p-orbital

D) It is electrophilic with sextet of electron

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q74

General Organic Chemistry – Compounds giving a positive bicarbonate test

JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 4

Top 100: **20%** got it right · Campus: **11%** (9 of 79)

Tough one

You: Skipped

74. How many of the following compounds can give positive bicarbonate test?

HCl, Picric Acid, Benzoic acid, 2,4-dintrophenol, 4-nitrophenol, Squaric Acid, formic Acid, Benzene Sulphuric acid, 1-Butyne, NH_4Cl .

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q75

General Organic Chemistry – Bases stronger than aniline

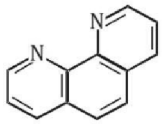
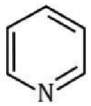
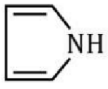
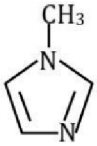
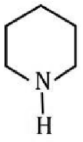
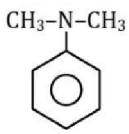
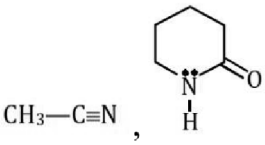
JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 5

Top 100: **11%** got it right · Campus: **9%** (7 of 79)

Tough one

You: Wrong

75. How many of the following are more basic then Aniline?

NH_3, $PhCH_2NH_2$,  ,  ,  ,  ,  ,  , $CH_3—CH_2—NH_2$, 

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

6 Every question at a glance

Right · Wrong · Skipped · Deleted (no marks) · Easy miss · Within reach · Tough one

Maths	1	2	**3**	**4**	5	6	7	**8**	9	10	11	**12**	13	14	15	16	**17**
Physics	**18**	**19**	**20**	21	22	**23**	**24**	25	**26**	**27**	28	**29**	**30**	**31**	**32**	**33**	**34**
Chemistry	35	**36**	37	38	39	40	41	42	43	44	~~45~~	**46**	**47**	48	49	**50**	**51**

7 Questions you lost – what to fix first

Every question you got wrong or skipped, one table per subject, easiest marks first. **Top 100** = the batch's top 100 across all campuses; **Campus** = all 78 students of this program on campus.

Easy miss 70%+ of the top 100 got it right – easiest marks to win back · **Within reach** 40–69% · **Tough one** below 40%

Maths

5 of 17 questions lost · **18** marks (+3 on partly-correct answers) · 2 within reach · 3 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q4	Within reach	**Functions**	Single correct · +3 / −1	Skipped	3	55%	65%
Q12	Within reach	**Functions**	Integer · +4 / 0	Wrong	4	55%	40%
Q3	Tough one	**Trigonometric Equations**	Single correct · +3 / −1	Wrong	4	36%	46%
Q8	Tough one	**Functions** – Period from two lines of symmetry	Integer · +4 / 0	Wrong	4	12%	13%
Q17	Tough one	**Trigonometric Equations** – Sum of solutions of $\cos(\pi\sqrt{(x-100)})\cdot\cos(\pi\sqrt{x}) = 1$	Paragraph · +3 / 0	Wrong	3	6%	6%

Physics

13 of 17 questions lost · **47** marks · 3 within reach · 10 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q23	Within reach	**Laws of Motion**	Multiple correct · +4 / −2	Skipped	4	56%	51%
Q31	Within reach	**Friction** – Friction on a block at rest inside an accelerating cube	Paragraph · +3 / 0	Skipped	3	45%	51%
Q19	Within reach	**Circular Motion**	Single correct · +3 / −1	Skipped	3	40%	38%
Q24	Tough one	**Circular Motion**	Multiple correct · +4 / −2	Skipped	4	34%	27%
Q20	Tough one	**Work, Energy & Power** – Maximum spring extension – energy method	Single correct · +3 / −1	Wrong	4	29%	27%
Q26	Tough one	**Circular Motion**	Integer · +4 / 0	Wrong	4	24%	17%
Q29	Tough one	**Circular Motion** – Net acceleration in a vertical circle at 37°	Integer · +4 / 0	Wrong	4	19%	14%
Q30	Tough one	**Friction** – Work by friction seen from an accelerating observer	Integer · +4 / 0	Wrong	4	19%	12%
Q18	Tough one	**Friction** – Static vs kinetic friction under a time-varying force	Single correct · +3 / −1	Wrong	4	16%	9%
Q34	Tough one	**Work, Energy & Power** – Work by conservative forces from the U–x graph	Paragraph · +3 / 0	Skipped	3	12%	9%
Q27	Tough one	**Friction** – Net work done by friction between sliding blocks	Integer · +4 / 0	Wrong	4	8%	10%
Q32	Tough one	**Friction** – Total force exerted by a block on an accelerating cube	Paragraph · +3 / 0	Skipped	3	8%	6%
Q33	Tough one	**Work, Energy & Power** – Work-energy theorem from a–t and U–x graphs	Paragraph · +3 / 0	Skipped	3	2%	1%

Chemistry

5 of 16 questions lost · 18 marks (+3 on partly-correct answers) · ● 1 within reach · ● 4 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q36	Within reach	**General Organic Chemistry**	Single correct · +3 / −1	**Wrong**	**4**	**64%**	53%
Q46	Tough one	**Isomerism** – Count of optically active compounds	Integer · +4 / 0	**Wrong**	**4**	**15%**	15%
Q51	Tough one	**Hydrocarbons** – Chiral products of free-radical monobromination	Paragraph · +3 / 0	**Wrong**	**3**	**15%**	4%
Q47	Tough one	**Hydrocarbons** – Dichlorination isomers of C_5H_{10} (VD 35)	Integer · +4 / 0	**Wrong**	**4**	**7%**	5%
Q50	Tough one	**Hydrocarbons** – Preparation of alkanes – count correct statements	Paragraph · +3 / 0	**Wrong**	**3**	**4%**	3%

8 Chapter scorecard

Your marks on each chapter's questions against the campus average for the same questions. Weakest chapters first within each subject. Chapter marks are approximate (≈): partly-correct answers cannot be traced to a question.

CHAPTER	QS	YOUR MARKS	YOU VS CAMPUS AVERAGE	CAMPUS AVG	QUESTIONS YOU LOST
Maths					
Functions	6	≈ **11**/22		40%	**Q4**, **Q8**, **Q12**
Trigonometric Equations	5	≈ **10**/17		35%	**Q3**, **Q17**
Inverse Trigonometric Functions	6	≈ **21**/21		55%	– none
Physics					
Friction	5	≈ **-1**/17		12%	**Q18**, **Q27**, **Q30**, **Q31**, **Q32**
Circular Motion	4	≈ **0**/15		17%	**Q19**, **Q24**, **Q26**, **Q29**
Work, Energy & Power	4	≈ **2**/12		22%	**Q20**, **Q33**, **Q34**
Laws of Motion	3	≈ **8**/12		39%	**Q23**
Momentum & Collisions	1	≈ **4**/4		33%	– none
Chemistry					
Hydrocarbons	5	≈ **8**/18		22%	**Q47**, **Q50**, **Q51**
Isomerism	2	≈ **4**/8		16%	**Q46**
General Organic Chemistry	9	≈ **26**/30		43%	**Q36**

Bar = your % of the chapter's marks | campus average

9 JEE Advanced · Maths – questions to revisit

5 questions · easy misses first
cropped from the question paper

Q4 — Functions

JEE Advanced · Maths · Single correct · marking +3 / −1 · marks lost 3
Top 100: **55%** got it right · Campus: **65%** (51 of 78)

Within reach
You: Skipped

4. Consider the function $f:N\to N, f(n)=p$ where 'p' is the largest prime number that divides n^2+n+41. Then f is :

A) one-one B) many-one C)onto D) into

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q12 — Functions

JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4
Top 100: **55%** got it right · Campus: **40%** (31 of 78)

Within reach
You: Wrong

12. Consider the function $f(x)=\log_{([x-1])}\left(\frac{|x|}{x}\right), ([.]=GIF)$

Let 'n' be the smallest positive integer in the domain of f. Let 'S' be the sum of all integers in the range of f. Then $n+S=$.....

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q3 — Trigonometric Equations

JEE Advanced · Maths · Single correct · marking +3 / −1 · marks lost 4
Top 100: **36%** got it right · Campus: **46%** (36 of 78)

Tough one
You: Wrong

3. If the equation $1+\sin^2(ax)=\cos x$ has a unique solution in 'x' then 'a' must be

A) rational B) irrational C) integer D) whole number

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q8

Functions – Period from two lines of symmetry

JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4

Top 100: **12%** got it right · Campus: **13%** (10 of 78)

Tough one

You: Wrong

8. Let $a \in R$. If $f(a-x) = f(a+x) \forall x \in R$, then $x = a$ is a line of symmetry in the graph of $y = f(x)$. Consider a non-constant function $g(x)$ such that $g(1+x) = g(1-x) \forall x \in R$ and $g(2026+x) = g(2026-x) \forall x \in R$. Then all such $g(x)$ must be periodic. The smallest positive number that could be a period for all such $g(x)$ is .....

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q17

Trigonometric Equations – Sum of solutions of cos(π√(x−100))·cos(π√x) = 1

JEE Advanced · Maths · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **6%** got it right · Campus: **6%** (5 of 78)

Tough one

You: Wrong

PARAGRAPH-II

Given positive integer 'k' , if we want to find positive integer 'n' and 'm' such that $n^2 - m^2 = k$ then we use $(n+m)(n-m) = k$ where we equate $n+m$ & $n-m$ to factors of k to find all possible values of (n,m).

Eg: $n^2 - m^2 = 6 \Rightarrow (n+m)(n-m) = (\pm 6)(\pm 1)$ or $(\pm 3)(\pm 2)$

17. The sum of real solution(s) of $\cos\left(\pi\sqrt{x-100}\right)\cos\left(\pi\sqrt{x}\right) = 1$ is .....

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q23

Laws of Motion
JEE Advanced · Physics · Multiple correct · marking +4 / −2 · marks lost 4
Top 100: **56%** got it right · Campus: **51%** (40 of 78)

Within reach
You: Skipped

23. An object of mass m is at rest in equilibrium at the origin. At $t = 0\text{sec}$ a force $\vec{F}(t)$ is applied which has the components $F_x(y) = k_1 + k_2 y$ $F_y(t) = k_3 t$ where k_1, k_2, k_3 are constants . Then

A) X-component of velocity at any time 't' is $V_x = \frac{k_1 t}{m} + \frac{k_2 k_3}{24 m^2} t^4$

B) X-component of acceleration at anytime 't' is $a_x = \left(\frac{k_1}{m} + \frac{k_2 k_3}{6 m^2} t^3 \right)$

C) Y- coordinate of object at any time 't' is $\frac{k_3 t^2}{3m}$

D) Y- coordinate of object at any time 't' is $\frac{k_3 t^3}{6m}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q31

Friction – Friction on a block at rest inside an accelerating cube

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

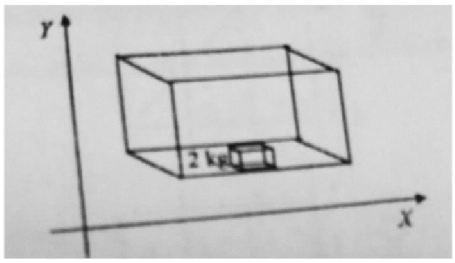
Top 100: **45%** got it right · Campus: **51%** (40 of 78)

Within reach

You: Skipped

PARAGRAPH-I

A solid block of mass 2 kg is resting inside a cube as shown in figure. The cube is moving with a velocity $\vec{V}=\left(5t\hat{i}+2t\hat{j}\right)m/s$. Here 't' is second. The block is in rest w.r.t cube and $\mu=0.5$ between block and surface of cube. $\left(g=10m/s^2\right)$



31. Force of friction acting on block is _____ N

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q19

Circular Motion

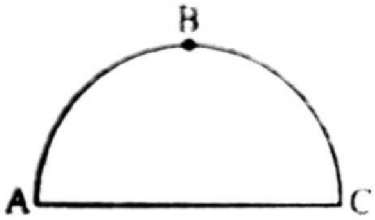
JEE Advanced · Physics · Single correct · marking +3 / −1 · marks lost 3

Top 100: **40%** got it right · Campus: **38%** (30 of 78)

Within reach

You: Skipped

19. If a particle starts from 'A' moving along the curved semi circular path as shown in figure with tangential acceleration 'a' then acceleration at B in magnitude is



A) $2a\left[\sqrt{1+\pi^2}\right]$ B) $a\left[\sqrt{1+\pi^2}\right]$ C) $a\left[\sqrt{\pi^2-1}\right]$ D) $a\pi\left[\sqrt{1+\pi^2}\right]$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q24

Circular Motion

JEE Advanced · Physics · Multiple correct · marking +4 / −2 · marks lost 4

Top 100: **34%** got it right · Campus: **27%** (21 of 78)

Tough one

You: Skipped

24. A particle is revolving in a circle of radius R with initial speed u. it starts retarding with constant retardation $\frac{u^2}{4\pi R}$. The number of revolutions it makes in a time $\frac{8\pi R}{u}$ is

A) 3 B) 2 C) 4 D) 6

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q20

Work, Energy & Power – Maximum spring extension – energy method

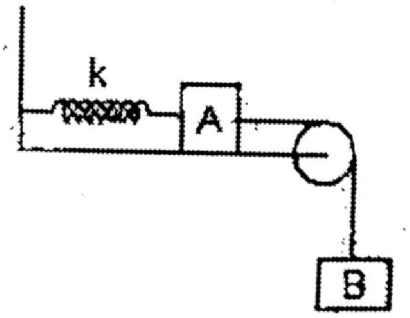
JEE Advanced · Physics · Single correct · marking +3 / −1 · marks lost 4

Top 100: **29%** got it right · Campus: **27%** (21 of 78)

Tough one

You: Wrong

20. All the surfaces are smooth system is released from rest with spring unstretched. Then maximum extension in spring will be $(m_A = m; m_B = 2m)$



A) $\frac{mg}{K}$ B) $\frac{2mg}{K}$ C) $\frac{3mg}{K}$ D) $\frac{4mg}{K}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q26 Circular Motion

JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **24%** got it right · Campus: **17%** (13 of 78)

Tough one

You: Wrong

26. A particle of mass 2.5kg is moving in gravity free space with velocity $3\hat{i}m/s$. At $t=0\text{sec}$ a force of magnitude 67.5N starts acting on the particle such that it is always perpendicular to instantaneous velocity. The minimum time after which particle has same velocity as the initial is $\frac{2\pi}{x}$ sec. The value of x ……..

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q29 Circular Motion – Net acceleration in a vertical circle at 37°

JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **19%** got it right · Campus: **14%** (11 of 78)

Tough one

You: Wrong

29. A small ball of mass m is attached to the end of string of length $l=1m$ whose other end is fixed. From its lowest position, the ball is given a kinetic energy $\frac{mgl}{5}$. Net acceleration $\left(in\ m/s^2\right)$ of ball at the instant when string makes an angle $\theta=37^{o}$ with vertical is ……$\left(g=10m/s^2\right)$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q30

Friction – Work by friction seen from an accelerating observer

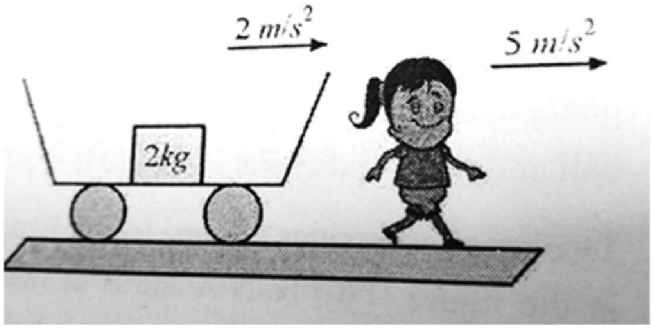
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **19%** got it right · Campus: **12%** (9 of 78)

Tough one

You: Wrong

30. An observer and a vehicle, both starts moving together from rest with acceleration $5m/s^2$ and $2m/s^2$ respectively. There is a 2 kg block on the floor of the vehicle and $\mu = 0.3$ between their surfaces. The work done by the frictional force on 2kg block as observed by running observer , during first 2seconds of motion is $\underline{-x}$ (J) $\left(g = 10m/s^2\right)$ find 'x'



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q18

Friction – Static vs kinetic friction under a time-varying force

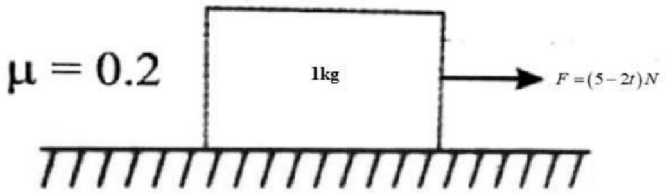
JEE Advanced · Physics · Single correct · marking +3 / −1 · marks lost 4

Top 100: **16%** got it right · Campus: **9%** (7 of 78)

Tough one

You: Wrong

18. The force acting on the block of mass 1 kg is given by $F = 5 - 2t$. The frictional force acting on the block after time $t = 2\text{sec}$ will be $\left(\mu = 0.2, g = 10m/s^2\right)$



A) 2N B) 3N C) 1N D) zero

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q34

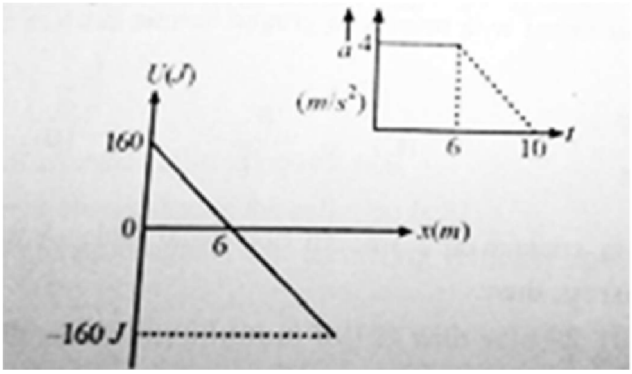
Work, Energy & Power – **Work by conservative forces from the U-x graph**
JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3
Top 100: **12%** got it right · Campus: **9%** (7 of 78)

Tough one
You: Skipped

PARAGRAPH-II

A particle of mass 1kg is moving along x-axis its velocity is 6m/s at x=0. Acceleration versus time and P.E versus displacement of particle are shown $\left(g = 10 m/s^2\right)$

Acceleration a is in $m/\sec^2$ b



34. Work done by conservative forces _____ J

Why did I lose this question? (write the reason and the correct approach)

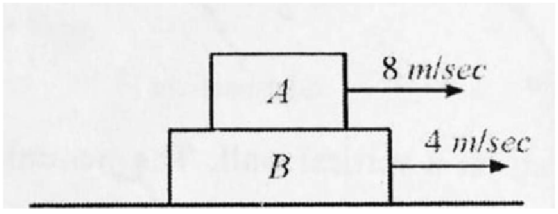
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q27

Friction – **Net work done by friction between sliding blocks**
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4
Top 100: **8%** got it right · Campus: **10%** (8 of 78)

Tough one
You: Wrong

27. Block A of mass 1kg is placed on rough surface of block B of mass 3kg. Block B is placed on smooth horizontal surface. Blocks are given velocities as shown. If net workdone by frictional force is "-n" J. value of n is ……..



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q32

Friction – Total force exerted by a block on an accelerating cube

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

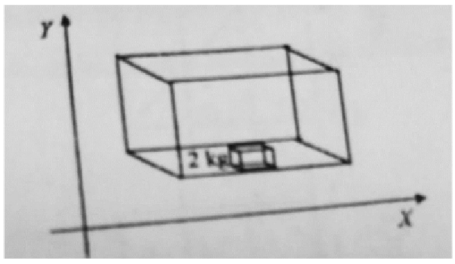
Top 100: **8%** got it right · Campus: **6%** (5 of 78)

Tough one

You: Skipped

PARAGRAPH-I

A solid block of mass 2 kg is resting inside a cube as shown in figure. The cube is moving with a velocity $\vec{V} = \left(5t\hat{i} + 2t\hat{j}\right) m/s$. Here 't' is second. The block is in rest w.r.t cube and $\mu = 0.5$ between block and surface of cube. $\left(g = 10 m/s^2\right)$



32. Total force exerted by block on cube is _____ N

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q33

Work, Energy & Power – Work-energy theorem from a-t and U-x graphs

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **2%** got it right · Campus: **1%** (1 of 78)

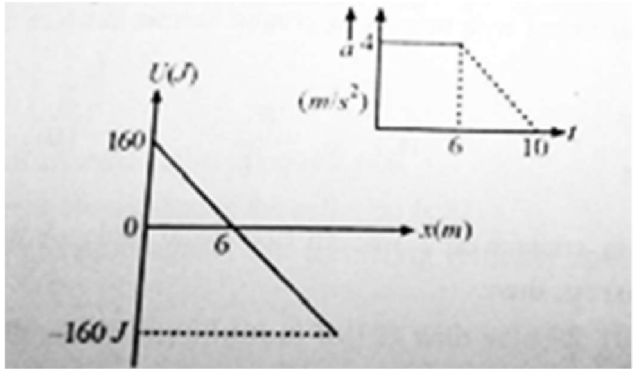
Tough one

You: Skipped

PARAGRAPH-II

A particle of mass 1kg is moving along x-axis its velocity is 6m/s at x=0. Acceleration versus time and P.E versus displacement of particle are shown $\left(g = 10 m/s^2\right)$

Acceleration a is in $m/\sec^2$ b



33. Work done by all forces _____ J

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q36

General Organic Chemistry

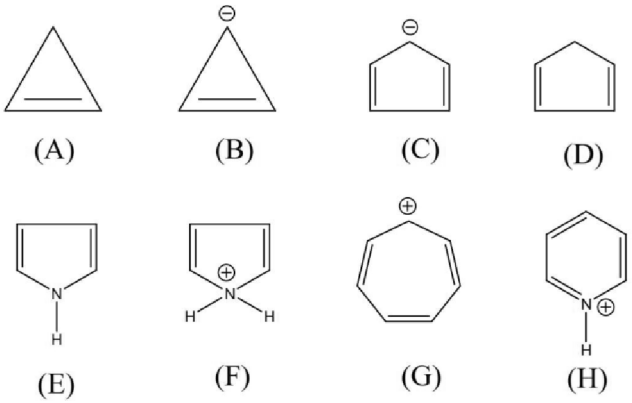
JEE Advanced · Chemistry · Single correct · marking +3 / −1 · marks lost 4

Top 100: **64%** got it right · Campus: **53%** (41 of 78)

Within reach

You: Wrong

36. Designate whether each of the following compound is aromatic (or) not aromatic



A) A,E,G,H are aromatic and B,C,D,F are not aromatic

B) A,C,E,G,H are aromatic and B,D F are not aromatic

C) A,C,D,E,G are aromatic and B,F,H are not aromatic

C) C,E,G,H are aromatic and A,B,D,F are not aromatic

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q46

Isomerism – Count of optically active compounds

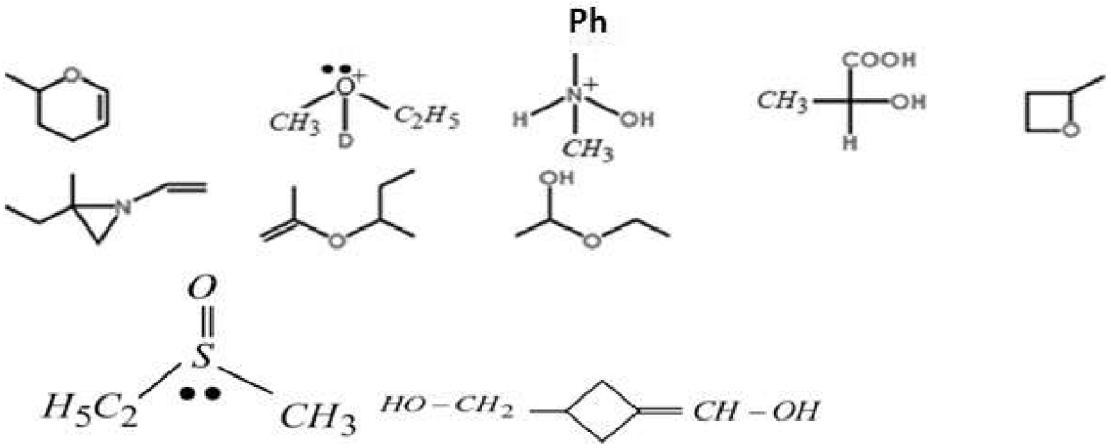
JEE Advanced · Chemistry · Integer · marking +4 / 0 · marks lost 4

Top 100: **15%** got it right · Campus: **15%** (12 of 78)

Tough one

You: Wrong

46. How many given compounds are optically active



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q51

Hydrocarbons – Chiral products of free-radical monobromination

JEE Advanced · Chemistry · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **15%** got it right · Campus: **4%** (3 of 78)

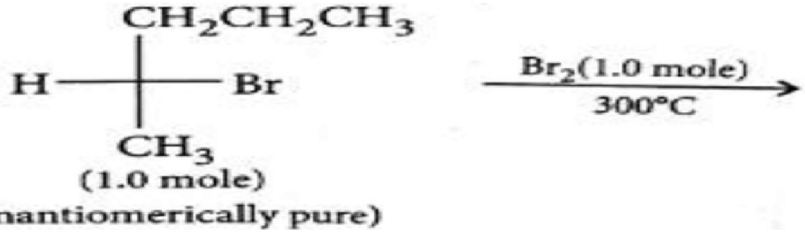
Tough one

You: Wrong

PARAGRAPH-II

Alkanes are aliphatic saturated hydrocarbons and have the general molecular formula C_nH_{2n+2}. The simplest one is methane (CH_4), is also the most abundant. Large amounts are present in atmosphere, in the ground, and in the oceans. Alkanes are most reduced forms of organic compounds. Hence, they can be prepared by reduction of any organic compound. Alkanes are prepared by the catalytic hydrogenation of alkenes or alkynes.

51. In the following monobromination reaction, the number of possible chiral products is



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q47 **Hydrocarbons – Dichlorination isomers of C_5H_{10} (VD 35)**

JEE Advanced · Chemistry · Integer · marking +4 / 0 · marks lost 4

Top 100: **7%** got it right · Campus: **5%** (4 of 78)

Tough one

You: Wrong

47. A hydrocarbon [X] is having vapour density 35 and it forms only 1 monochloro product under sunlight. How many isomeric products (including stereo) are possible when [X] undergoes dichlorination?

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

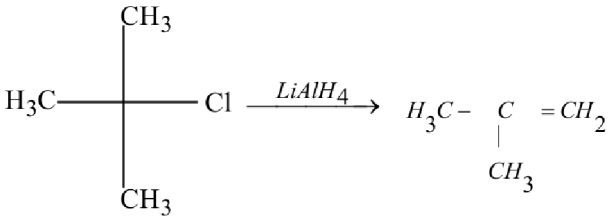
PARAGRAPH-II

Alkanes are aliphatic saturated hydrocarbons and have the general molecular formula C_nH_{2n+2}. The simplest one is methane (CH_4), is also the most abundant. Large amounts are present in atmosphere, in the ground, and in the oceans. Alkanes are most reduced forms of organic compounds. Hence, they can be prepared by reduction of any organic compound. Alkanes are prepared by the catalytic hydrogenation of alkenes or alkynes.

50. Find the No of correct statements out of the following

A) Alkanes are prepared by catalytic hydrogenation of Alkenes where both hydrogen atoms add to the same face of the alkene (syn-addition).

B) Alkyl halides when reacts with Lithium Aluminium Hydride form corresponding alkane: $R-X \xrightarrow{LiAlH_4} R-H$ for R : methyl , $1^0, 2^0$

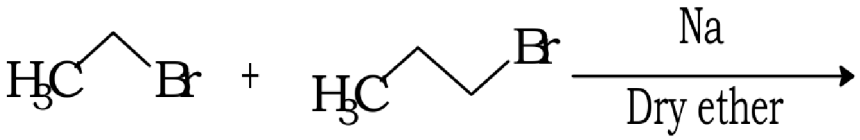


C)In case of Tertiary Alkyl halide is observed.

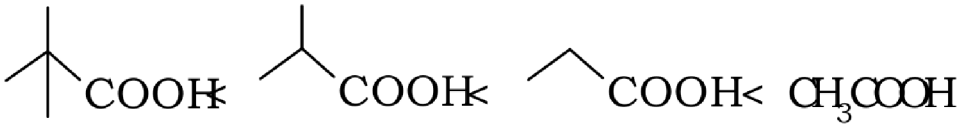
D) Grignard's reagent when treated with a polar protic solvent gives alkane:

$R-Mg+HA \rightarrow R-H+MgX(A)$

E) The different compounds obtained by the following reaction are 7 (only organic compounds)



F) Rate of sodalime decarboxylation for the following acids follow:



G) The side products of potassium butanoate during Kolbe's Electrolysis are propane, propene and propyl butanoate

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

10 How to read this report

Where the numbers come from. Marks, percentages, All-India / State / Campus ranks, subject marks, subject All-India ranks and the wrong / skipped questions are the official results in the institute's *All India Marks Analysis* (graded with the final answer key). Maximum marks leave out deleted questions — JEE Mains: Q58 — JEE Advanced: Q45. **Program rank** = rank in Pragya G9 (EXT) Achievers across all campuses, from the batch's question-wise sheet († JEE Advanced: that sheet was graded before the final key, so this rank may differ slightly). **Top 100** = the batch's top 100 across all campuses, from the same question-wise sheet; * where it has no data for a question, the campus figure decides the priority. **Campus** = all students of this program on campus. **Priority:** Easy miss = 70%+ of the top 100 got it right; Within reach = 40–69%; Tough one = below 40%. **Marks lost** on a question = its marks plus the negative mark if answered wrong, its marks if skipped. In JEE Advanced, 6 marks went on multiple-correct answers that were only partly correct. **Weekly-test average** = your average over the weekly Mains tests (WTM) or Advanced tests (WTA) before this Part Test, from the campus averages sheet. The weekly Advanced tests are marked out of 240; their averages are scaled to 180 to match this paper (× 180/240, so 80 per subject becomes 60). Percentages are unchanged.